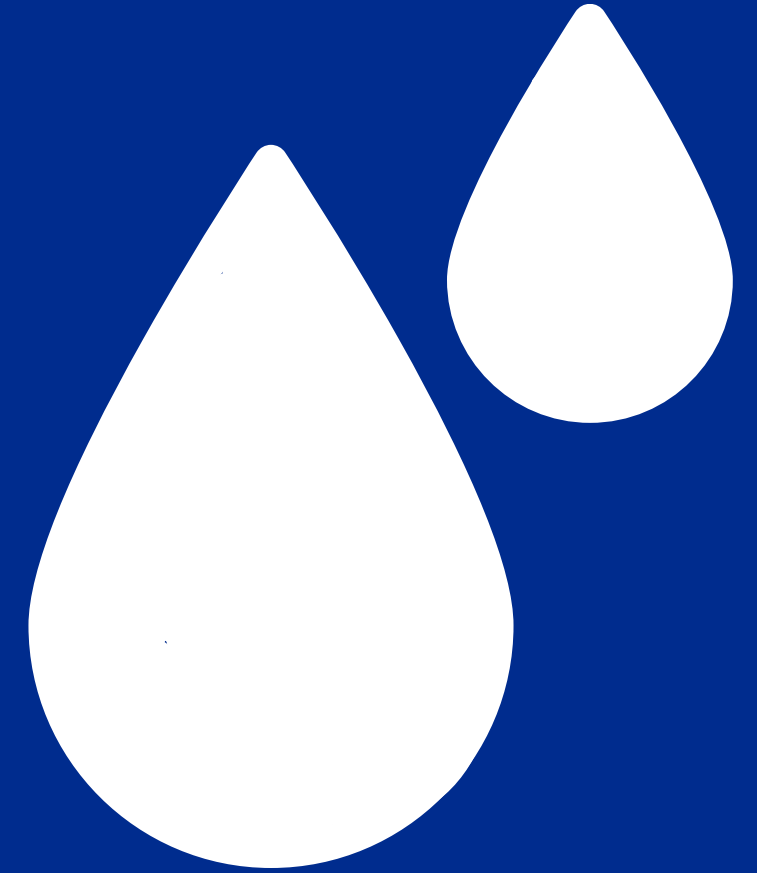


**Jesus & Mary College
Delhi University**

Team: CASE_RUNNERS

Presented by :

**HEMANT KUMAR
AADITYA GUPTA
DHRUV MAHICH**



The Last 10% Problem in Personal Care Packaging

Understanding the industry and the problem.

The Problem

Consumers consistently lose 10–15% of product value in products like Nivea Body Lotion Dispenser Bottles due to inefficient packaging which leads to frustration, wasted value and unmet expectations of “*Paisa Vasool*.”

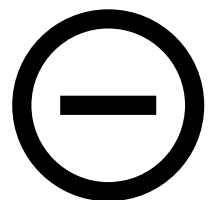
Objectives



Maximize Product Utilization i.e. ensure near-total usage of the product.



Improve Value Perception & Sustainability of the product.



Remove the need for shaking, cutting or “Bottle Surgery”.



Commercially viable and culturally relevant solution.



Core Insight and Existing Issues

Inefficient packaging causes consumers to consistently lose upto 15% of the product requiring manual efforts i.e. “Bottle Surgery” at end

STANDARD DIP PUMP – HOW IT WORKS

PUMP HEAD
User presses to dispense product

PUMP MECHANISM
Creates suction to pull product up

COLLAR
Secures the pump to the bottle

DIP TUBE
Reaches down to the bottom to draw product

BOTTLE
Holds the product

STEP-BY-STEP WORKING

1 Press

User presses the pump head.

2 Suction Created

The mechanism creates a vacuum inside.

3 Product Drawn Up

Lotion is pulled up through the dip tube.

4 Dispensed

Lotion flows out of the nozzle.

THE LIMITATION

When the lotion level drops below the dip tube opening, the pump can no longer draw product, leaving ~10–15% of lotion trapped at the bottom and on the walls.

Unused product (waste)

MAJOR GAPS	Functional Gap	Pump fails before product ends	
	Usability Gap	“Bottle Surgery” is required for full product recovery	
	Hygiene Gap	Exposure to air & contamination	

Residual Waste

- Upto 15% of the lotion remains trapped inside the bottle at end.
- Once the lotion level drops below the tube opening, the pump starts pulling air instead of product.

Frustrating End Experience

- Users struggle at the final stage when pump stops working.
- Shaking or cutting the bottle becomes necessary to extract the remaining product.

Environmental Impact & Value Perception

- Residual chemicals in discarded bottles contribute to waste and pollution.
- Consumers feel shortchanged—paying for more than they can use.

Our Design and Innovation

We shift from a Pull-based system (failure-prone) to a Push-based system (near-total evacuation)

OUR INNOVATION

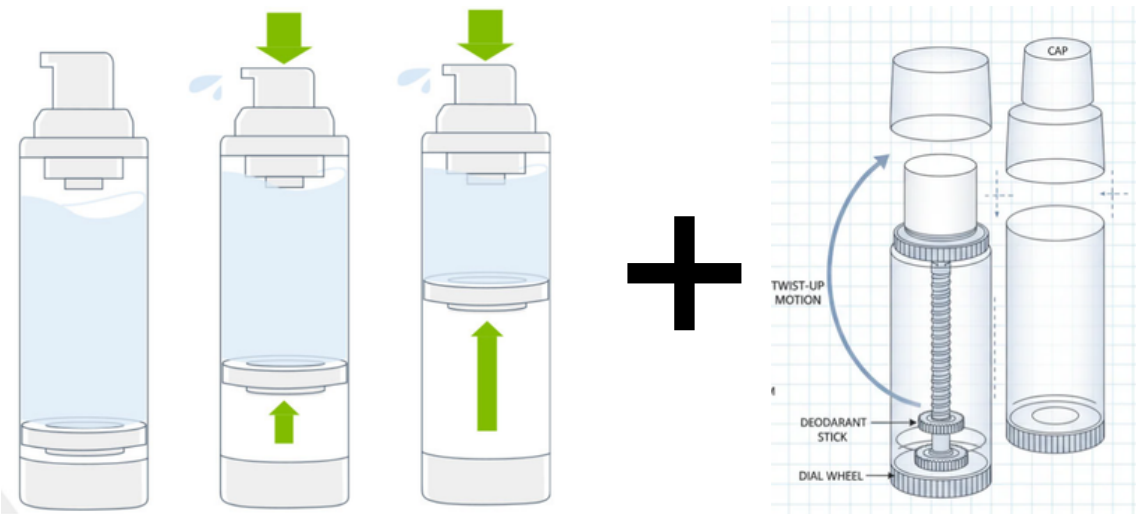


Core Design

- **Airless Pump Mechanism** → Eliminates dip tube entirely
- **LiftBase (Piston)** → Pushes the product upward as the piston rises
- **WipeFlow Ring (Silicon)** → Removes side-wall residue and redirects it to usable flow.
- **LockFlow Ratchet** → Manual system, one-way base movement similar to what's seen in Lipsticks with a lock.

Proposed Solution

Relatively similar to the system shown below



Working

How the System Works

- User presses pump → lotion dispensed normally
- Pump action creates a small internal vacuum
- Product rests on piston at the bottom
- Vacuum allows piston to move slightly upward
- Lotion is pushed toward the nozzle for continuous flow

End-of-Use Mechanism (LockFlow)

- When pump efficiency reduces, user can manually twist the base
- Ratchet mechanism pushes piston upward in fixed steps
- Ensures one-way movement (no backward slipping) with a lock
- Similar to lipstick-style mechanism

Efficiency Enhancement

- Wiper ring scrapes sidewall residue
- Redirects leftover lotion into usable flow
- Ensures near-total product utilization (~98%)

Impact - From Wastage to Value Creation

Transforms packaging into a value delivery system



Consumer Impact

- **True “Paisa Vasool”** – Consumers feel they get full value for what they paid (No lotion left at the bottom or sidewalls i.e. complete product usage ~98%)
- **Higher Value for Money** – ~10% more usable product, translating to ~₹40 extra value
- **Hassle-Free Experience (No Bottle Surgery)** – No need to shake, cut, or struggle with the bottle without mess



Brand Impact

- **Higher Perceived Value** – Delivers more usable product without changing formulation
- **Stronger Customer Loyalty** – Solves a common pain point, increasing repeat purchases
- **Premium & Sustainable Positioning** – Differentiates the brand while reducing chemical waste (Positions the brand as innovative and environmentally responsible)



Financial Impact

- Traditional dip-tube system wastes ~48 ml (~₹48 value lost per bottle)
- LockFlow system reduces waste to ~8 ml, enabling near-total usage
- Net recovery of 40 ml delivers ~₹40 additional consumer value
- Also enables the option for price increase & margin expansion for the brand

Metric	Traditional System (Dip-Tube)	Proposed Design (LockFlow System)	Key Impact
Total Volume	400 ml	400 ml	No change in formulation cost for brand
Product Waste (%)	~12% (On Average)	~2%	↓ 10% Waste (lesser chemical residue)
Volume Wasted (ml)	~48 ml	~8 ml	~40 ml Recovered by the consumer
Usable Product (ml)	~352 ml	~392 ml	+40 ml Usable product for the consumer
Value of Wasted Product	₹48	₹8	₹40 Reduction
Consumer Value Extracted	₹352 worth (₹1/ml basis)	₹392 worth	+₹40 Value gained
Net Value Gain	-₹40 (value lost by consumer)	+₹40 (value recovered)	Converts loss into gain

Business Case and Unit Economics

A balanced view of value creation, execution risks, and economic viability of the LockFlow system

Cost Component	Existing Product (Dip-Tube System)	LockFlow System (Proposed)	Delta (₹)
Formulation + Filling	₹250–₹280	₹250–₹280	0
Bottle Body (PET/HDPE)	₹10–₹12	₹12–₹15 (slightly reinforced)	+2–3
Standard Pump Mechanism	₹8–₹10	—	–8–10
Airless Pump System	—	₹12–₹15	+12–15
Dip Tube	₹1–₹2	—	–1–2
Piston Base (LiftBase)	—	₹3–₹4	+3–4
Wiper Ring (Silicone)	—	₹2–₹3	+2–3
Ratchet Lock (LockFlow)	—	₹3–₹4	+3–4
Assembly & Manufacturing	₹2–₹3	₹4–₹6	+2–3
Total Packaging Cost	₹20–₹30	₹40–₹60	+₹15–₹20
Total Product Cost	₹270–₹310	₹290–₹330	+₹15–₹20



Potential Risks

- **Pricing Sensitivity:** Mitigated by targeting premium segments and anchoring on clear value gain (₹40 benefit vs ₹20 cost)
- **Consumer Adoption:** Minimal behavior change by retaining familiar pump usage; simple manual assist only at end-of-use
- **Mechanical Reliability:** Use of proven airless and ratchet systems with rigorous durability and lifecycle testing

“All key risks are mitigated through proven technologies, minimal behavior change, and strong value-led positioning”

Sustainability, Feasibility & Functionality

Simple to use, easy to scale, better for environment



Sustainability

- **Net Reduction in Environmental Waste:**

Decreases product residue from ~12% to ~2%, ensuring significantly less chemical waste enters landfills and water systems

- **Improved End-of-Life Recyclability:**

Near-complete product evacuation leaves bottles cleaner, making them more compatible with existing recycling processes

- **Material vs Waste Trade-off Optimized:**

While packaging material increases marginally, it is offset by a substantially higher reduction in wasted product—resulting in a net positive lifecycle impact

- **Built on Existing, Scalable Materials:**

Uses standard recyclable materials (PET/HDPE) and proven components, avoiding introduction of complex or non-recyclable elements



Feasibility

- **Proven, Industry-Standard Components:**

Leverages widely used airless pumps and ratchet mechanisms, eliminating technology uncertainty and reducing development risk

- **Minimal Manufacturing Disruption:**

Compatible with existing PET/HDPE molding and assembly lines, using proven materials (PET/HDPE body, silicone wiper, mechanical ratchet)—no new technology required, only smart integration

- **High Reliability, Low Failure Risk:**

Purely mechanical system with no electronics, supported by established durability benchmarks from similar applications

- **Scalable & Cost-Controlled Deployment:**

Modular design enables easy scaling across SKUs, with standardized components allowing efficient sourcing and cost optimization



Functionality

- **Airless, Push-Based Dispensing:**

Replaces dip-tube with a piston system that ensures consistent, uninterrupted product flow which is already used in several products

- **No Learning Curve Required:**

Maintains familiar pump usage, enabling seamless adoption without behavioral change

- **Minimal Effort at End-of-Use:**

Simple manual twist activates LockFlow, requiring low effort and no external tools relatively similar to Lipsticks as seen in the cosmetics industry

- **Reliable, Controlled Performance:**

Vacuum-assisted system with ratchet lock prevents backflow, ensuring precision and consistency

THANK YOU
